

High-Capacity QLC Innovation

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DDR reduction on high-capacity SSD

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Exploring DDR reduction on high-capacity SSD

- High capacity QLC SSD typically start at 32 TB (raw) and go up to 128 TB or 256TB. The industry is now starting to look at producing 512 TB SSD.
- L2P caching in DRAM typically follows either a 1000:1 (4K IU, low-capacity SSD) or 4000:1 (16K IU, high-capacity SSD), but this still requires a lot of RAM.
  - Many designs are limited to 2 ranks to manage speed vs loading
  - There is a practical ceiling of 128 GB without adding more pins for wider bus

Observations

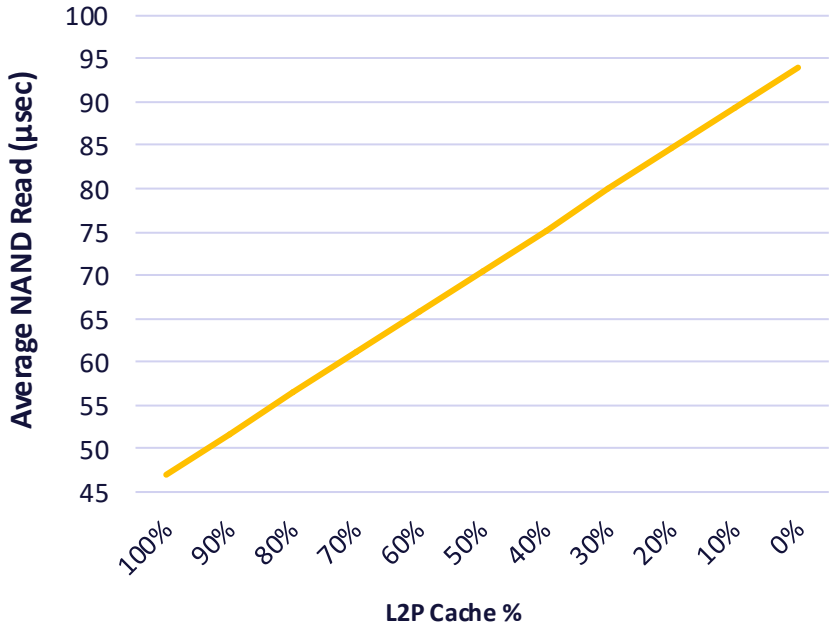
- Taken as a percentage, the BOM Cost of DRAM is ~1%
- If you're buying 20 EB of SSD, this can represent $150 million in CAPEX, which could be used for something else
- Two paths for DDR reduction
  A. Reduce L2P caching in DRAM down from 100%
  B. Consider larger Indirection Units (IU)

SSD Capacity (TB)	SSD Price	L2P DDR @ 16K IU (GB)	DRAM Cost
32	$18,000	8	$230
64	$36,000	16	$461
128	$72,000	32	$922
256	$144,000	64	$1,845
512	$288,000	128	$3,690

Note: QLC Enterprise SSD $0.589 $/GB, DDR5 $28.83 $/GB

DDR Reduction: Path A – L2P Caching Reduction

- Looking only at NAND read latency, we can see a linear relationship between L2P cache size and access time.
- As the caching trends 100% → 0%, then the NAND reads for each LBA access also trends 1 → 2.
- This will show up as direct latency on every read and not a QoS value pushed out to five-nines (ie: QoS 99.999).



Note: Assuming 47 µsec per read command

Questions

1. Is this latency relevant to large scale deployments, or are other factors adding so much additional latency that +47 µsec per read doesn't matter?
2. Can we organize the data such that we can ensure locality and thus work with a much smaller L2P cache? Additionally, does this concept scale from 1 SSD to 100,000 SSD?

DDR Reduction: Path B – Larger L2P IU

- With larger IU, there is no need to reduce L2P caching, thus no impact on read latency
- We see there is no meaningful impact on WAF for AI Workloads up to 64K IU. We can compensate for GC WAF with slightly more OP
- We know that the standard JEDEC Enterprise Workload (table not shown) is highly penalized due to small writes pushing WAF from 2.5 up to 16.7. The only option to compensate for this huge WAF is using very high-endurance NAND.

Workload	Write pattern	4K	16K	32K	64K	128K	128K vs 4K
ResNet-50 datagen	300 files × ~134 MB	4.806	4.796	4.791	4.779	**4.787**	**±0%**
UNet3D datagen	260 files × ~146 MB	4.100	4.099	4.102	4.147	**4.189**	**+2%**
CosmoFlow datagen	14,700 files × ~2.7 MB	3.400	3.451	3.530	3.736	**4.194**	**+23%**
KV-cache offload	kv-cache.py, direct	2.682	2.620	2.826	2.951	**3.688***	**+37%**
RetinaNet datagen	48,500 files × 323 KB	1.637	1.745	1.886	2.088	**2.760**	**+69%**
DLRM datagen	10 parquet files	2.260	2.326	2.499	2.528	**3.390**	**+50%**
Flux datagen	25 parquet files	1.975	2.011	2.192	2.205	**3.049**	**+54%**
VectorDB datagen	Milvus DiskANN	1.581	1.621	1.810	1.844	**2.766**	**+75%**
VectorDB run	Queries + compaction	1.236	1.240	1.417	1.354	**2.263**	**+83%**
Checkpoint	llama3-8b, 1 × 105 GiB	1.022	1.020	1.044	1.060	**1.290**	**+26%**
Weighted (9 stages)	**excl. VectorDB run**	**2.449**	**2.462**	**2.529**	**2.628**	**2.980**	**+22%**

DDR Reduction: Conclusion

- Holding to one-size-fits-all solutions limits options for BOM cost reduction
- Looking at scaling drives based on target workloads helps, particularly for hyperscalers
- Brining back the example for the first slide, at 20EB of SSD, the cost for DRAM goes from $150 million → $37 million
- This represents a 75% DDR BOM reduction or $113 million that can be used for something else

SSD Capacity (TB)	SSD Price	L2P DDR @ 16K IU (GB)	DRAM Cost @ 16K IU	L2P DDR @ 64K IU (GB)	DRAM Cost @ 64K IU
32	$18,000	8	$230	2	$57
64	$36,000	16	$461	4	$115
128	$72,000	32	$922	8	$230
256	$144,000	64	$1,845	16	$461
512	$288,000	128	$3,690	32	$922

Note: QLC Enterprise SSD $0.589 $/GB, DDR5 $28.83 $/GB

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Reducing DRAM Use in QLC SSDs

Rory Bolt, Senior Fellow, KIOXIA America, Inc.

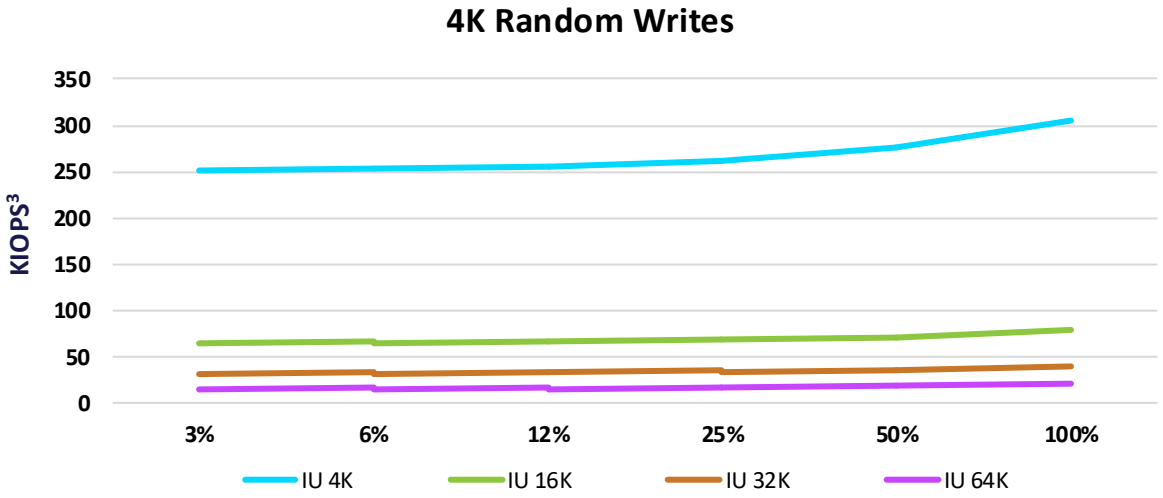
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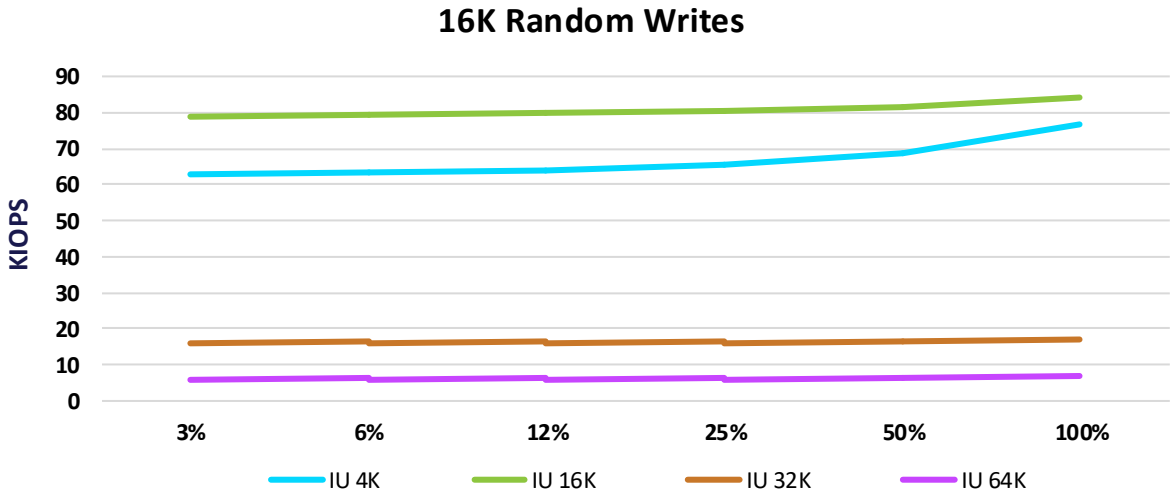
Overview

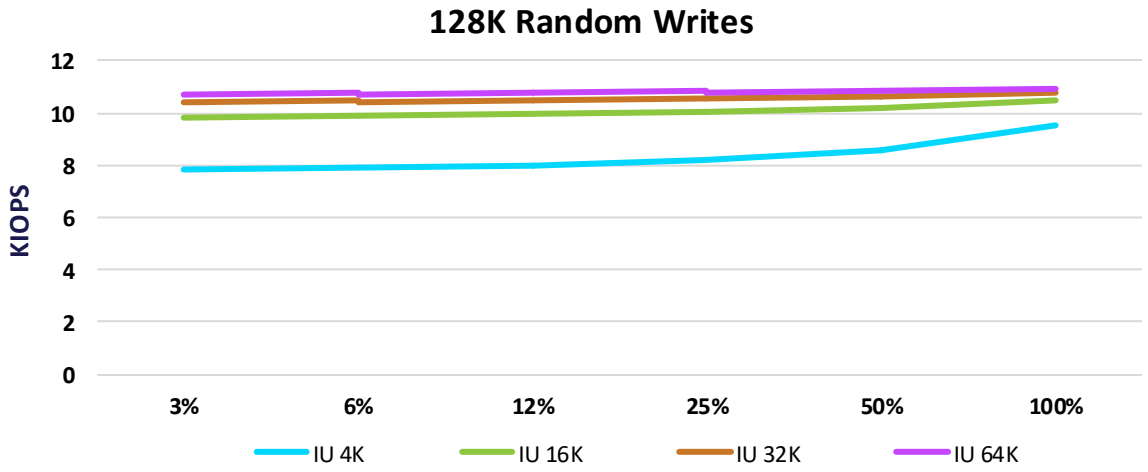
- It is desirable to limit DRAM capacity of large data storage drives:
  - Cost/supply
  - Physical space
- In data center drives, most DRAM is used by the Look Up Table (LUT)
- Two different approaches to reduce DRAM size are compared:
  - LUT caching
  - Large Indirection Unit (IU) size
- This study:
  - Focuses on random write performance of a hypothetical QLC drive
  - Uses simulation/calculation for rough performance estimation
- Theoretical limits are presented, overheads may result in lower performance

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Hypothetical 128 Terabyte[1] QLC SSD Performance[2]







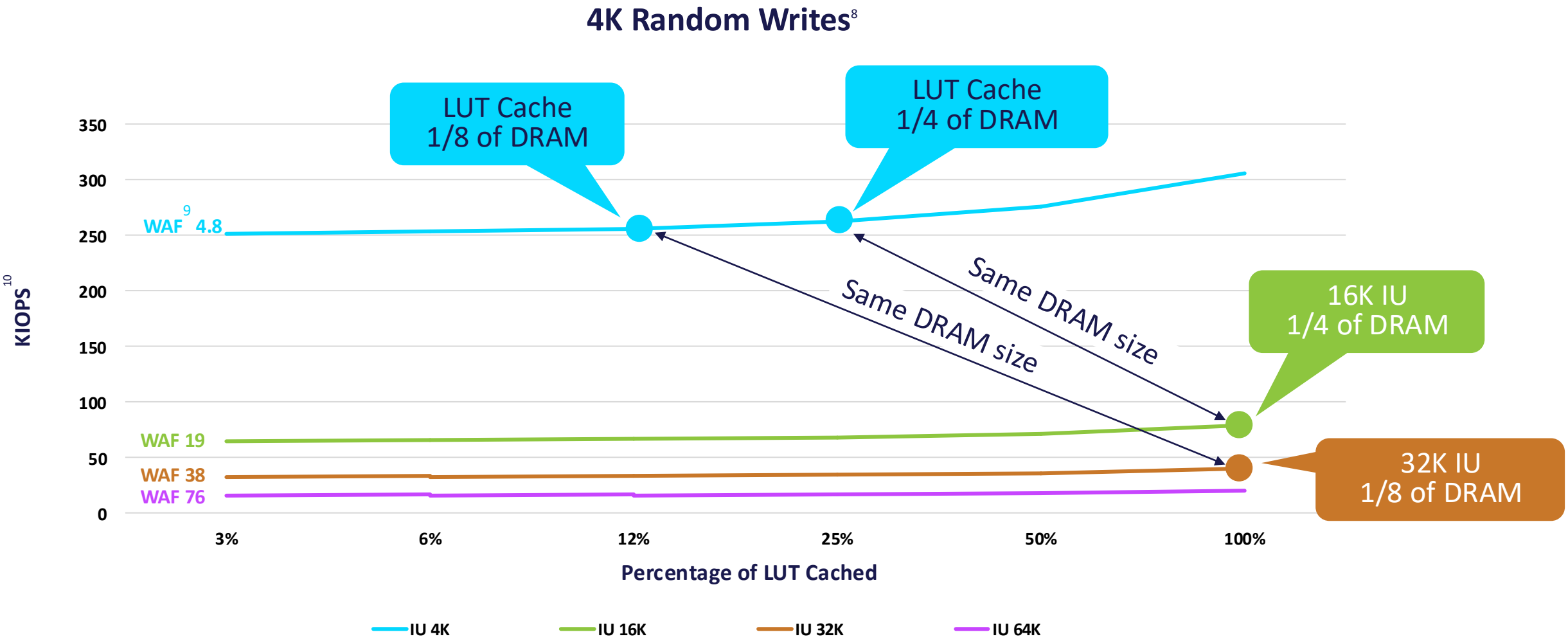
- 100% random write workload
- Horizontal axis is percentage of LUT[4] cached
- If IU[5] > access size, performance is reduced
- LUT cache performance is consistent from very low percent to 100%
- If IU = access size, performance is optimal

1. Definition of capacity: KIOXIA Corporation defines a megabyte (MB) as 1,000,000 bytes, a gigabyte (GB) as 1,000,000,000 bytes, a terabyte (TB) as 1,000,000,000,000 bytes and a petabyte (PB) as 1,000,000,000,000,000 bytes. A computer operating system, however, reports storage capacity using powers of 2 for the definition of 1Gbit = 230 bits = 1,073,741,824 bits, 1GB = 230 bytes = 1,073,741,824 bytes, 1TB = 240 bytes = 1,099,511,627,776 bytes and 1PB = 240 bytes = 1,125,899,906,842,624 bytes and therefore shows less storage capacity. Actual formatted capacity may vary.

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Observation 1: LUT[6] Caching is Effective for Random Write Size < IU[7] Size

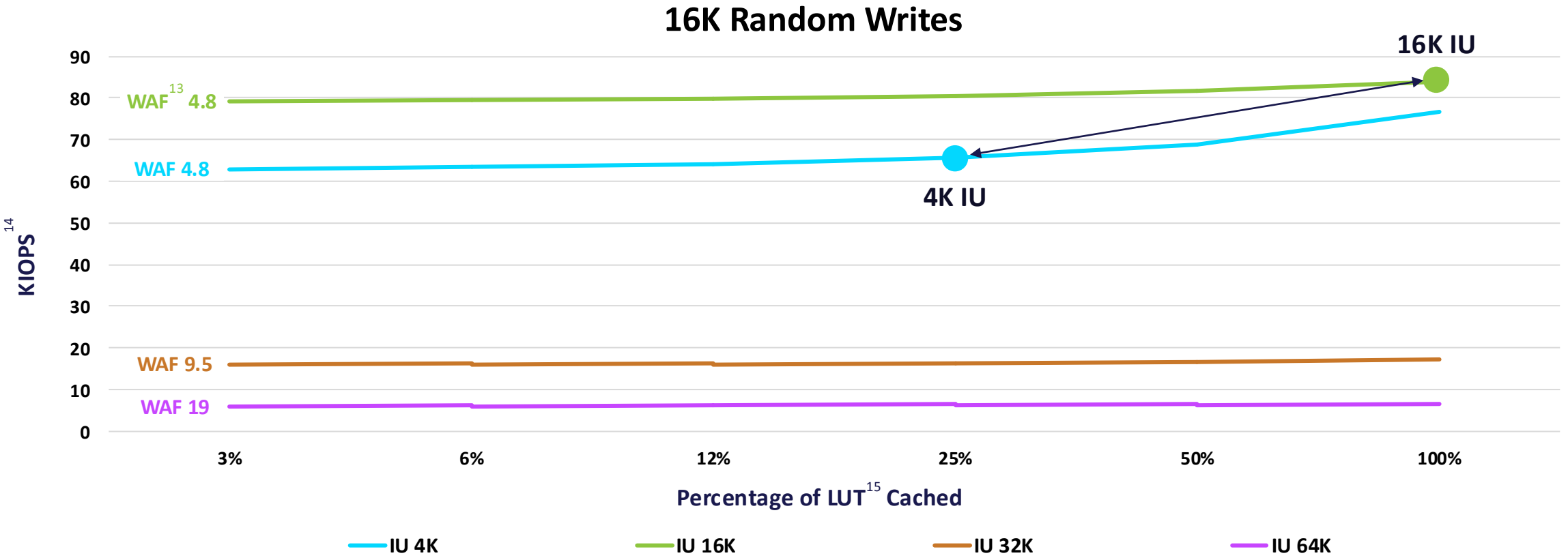


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Observation 2: Smaller IU[11] Size May Not Always Win

16K IU performance is better than 4K IU performance for 16K random writes[12]

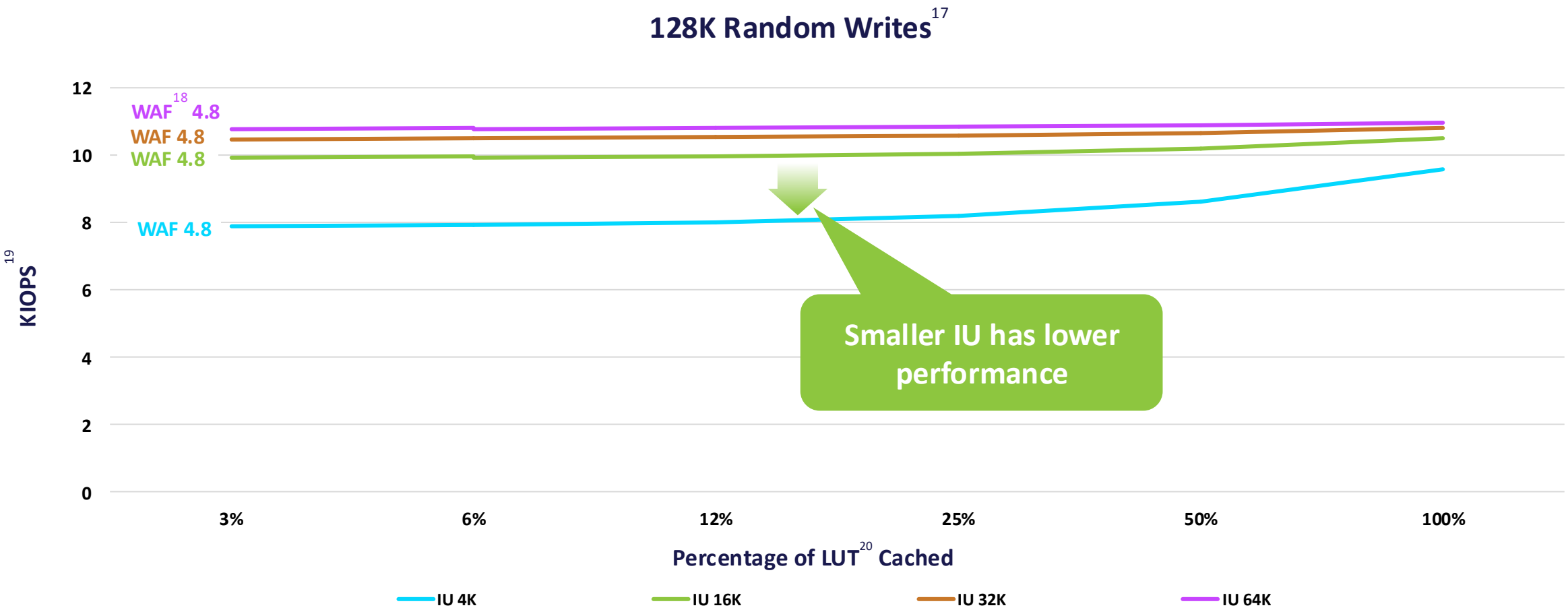


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Observation 3: Larger IU[16] is Better in Large Access Size Random Write Workload

Performance: IU 4K < IU 16K < IU 32K < IU 64K



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Scaling QLC to 100+ TB: Three Controller Challenges

Yang Liu, VP of ASIC Engineering, ScaleFlux

100+ TB QLC: Three Challenges

1 · FTL metadata
4 KB map ≈ 1 GB DRAM per TB — ~128 GB at 128 TB. The map outgrows DRAM.

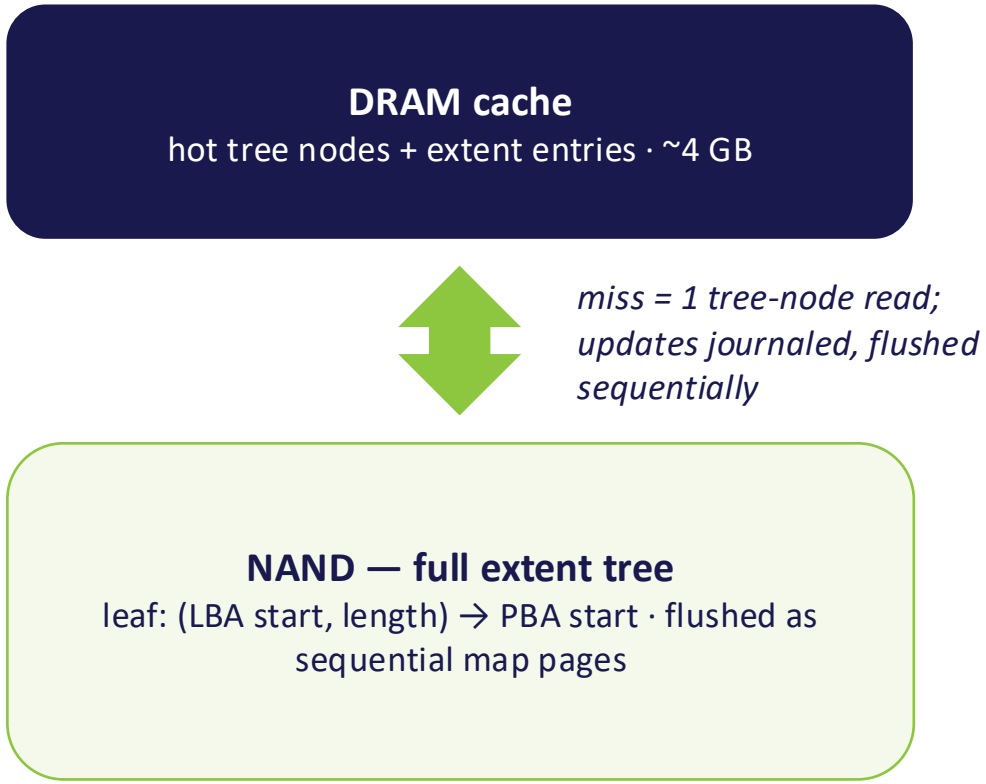
2 · Write efficiency
Few-hundred P/E cycles — GC copies and misplaced writes burn endurance QLC doesn't have.

3 · Failure domain
One drive holds a shelf's worth — die failures and rebuilds scale with capacity.

Drive capacity	32 TB	64 TB	128 TB	256 TB
FTL DRAM needed (1 GB/TB)	32 GB	64 GB	128 GB	256 GB
DDR5 modules (4 GB each)	8	16	32	64

All three are controller problems: adaptive hierarchical FTL · host placement (FDP) · dual-parity internal protection

Adaptive Hierarchical FTL

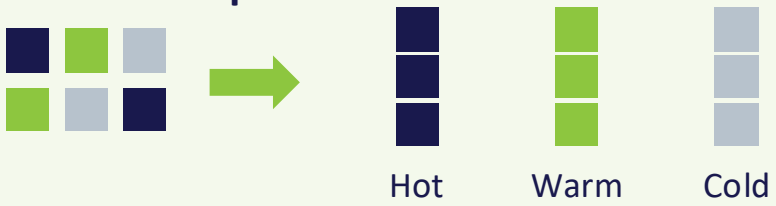


- **Tree, not array:** map is a B+-tree of extents — one entry spans any run of consecutive LBAs → consecutive PBAs (start, length, target)
- **Adaptive:** a 1 MB sequential write needs 1 entry vs. 256 page entries — the table shrinks as workloads write larger chunks; 4 KB randoms fall back to page-size extents
- **Hierarchical:** full tree persists in NAND; DRAM demand-caches hot nodes (DFTL-style), updates journaled for power-loss safety
- **Nearline fit:** AI data-lake writes are large and sequential, reads dominate — few extents, and a rare miss costs one flash read

Payoff: map size tracks write pattern, not capacity — 122/245 TB-class QLC no longer needs ~1 GB DRAM per TB

Write Efficiency: FDP + Compression

FDP — data placement



Host tags writes by lifetime (NVMe / OCP standard) — like-lifetime data dies together, GC copies collapse

Transparent compression



Inline in the controller, invisible to the host — WA drops below 1× on compressible data

Coarse IUs and compressed extents land in the same L2P — co-designed with the hierarchical FTL

Write amp ↓ · GC copies ↓ · over-provisioning ↓ → more usable TB per drive

Containing the Failure Domain

The problem at 100+ TB

Blast radius
One drive = a full shelf of data

Noisier media
QLC: higher RBER, tighter margins

Slow recovery
Host-level rebuilds take days

The answer: 18 data channels + die-level RAID 6



2/18
die failures absorbed

100%
capacity retained (= 16 ch)

Plus die-level RAID 6: rebuilds run inside the drive at NAND bandwidth — a die failure is a background repair, invisible to the host

The bar: hold host-visible UBER flat at 100+ TB — while spending less ECC margin per cell

Three Challenges, Three Answers

Challenge at 100+ TB	Controller answer	What it buys
1 · FTL metadata Map outgrows DRAM: ~1 GB per TB	**Adaptive hierarchical FTL** Extent-tree map lives in NAND; DRAM caches hot nodes. Map size tracks write pattern, not capacity.	**~4 GB** DRAM at 128 TB, not ~128 GB
2 · Write efficiency Few-hundred P/E cycles; GC burns them	**FDP placement + compression** Like-lifetime data dies together; inline compression shrinks what remains. Both co-designed into the same L2P.	**< 1×** write amp on compressible data · less GC, less OP
3 · Failure domain One drive = a shelf; QLC is noisier media	**18 channels + die-level RAID 6** Two spare channels, dual parity across dies; rebuilds run inside the drive at NAND bandwidth.	**2/18** die failures absorbed · 100% capacity · UBER flat

Takeaway: at 100+ TB, DRAM cost, endurance and reliability are set by the controller — not by the NAND cell
Ask three numbers: DRAM per TB · WAF on your workload · die failures survived in-drive



High-Capacity QLC Innovation

Mark Dancho, Distinguished Firmware Engineer, Sandisk

High-Capacity SSD Innovation #1: SPRANDOM

➢Problem:

- SSD preconditioning time before starting test.
  - 128 TB SSD takes ~12 days
  - 256 TB SSD takes ~24 days
- Qualification and workload tests are practically non-viable.

How to accelerate qualification?

More Information:



Solution:

- A new open-source precondition methodology called **SPRANDOM**
- SPRANDOM methodology enables a 128 TB SSD to precondition in ~14 hours*

Note: Works on QLC, TLC, and all capacity SSDs

High-Capacity SSD Innovation #2: SSD Capacity Scaling

Problem:

- Need a form factor to scale capacity.

Form Factor	NAND Placements
E1.S	8
E3.S/L	16–32
E1.L	32

- Need more NAND placements to scale capacity



Solution:

E2 (SFF-TA-1042)

- 64 NAND Placements – enables scaling towards 1 PB

High-Capacity SSD Innovation #3: Efficient FTL

Traditional Datacenter Approach:
FTL is in DRAM enabling
high performance

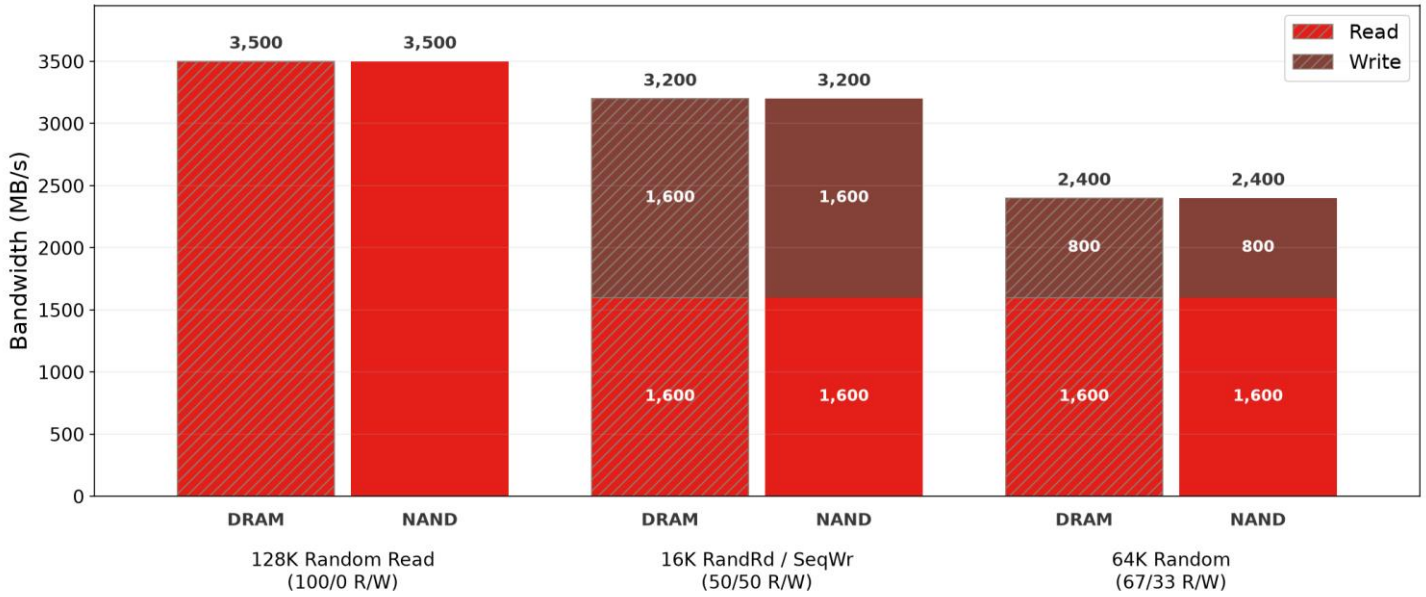
Traditional Client Approach:
FTL is in NAND with
reduced performance

New Datacenter High-Capacity
FTL Approach: FTL in NAND

FTL in NAND Performance Impact

❖Example High-Capacity / Rate Limited Workloads:

- 128K Random Read
- 16K Random Read / Sequential Write 50/50
- 64K Random 67/33 Read/Write



FTL in NAND and FTL in DRAM achieved comparable results across tested workloads.

Efficient QLC High-Capacity FTL Example – Illustrative Model

Example Deployment Plan:

- Plan to deploy 1 EB of useable capacity
- System
  - 96 GB DRAM
  - 24 SSDs
  - SSD Information
    - 128 TB Raw with 122.88 TB useable capacity
    - 16K IU size
      - Option 1 SSD with Traditional FTL: 32 GB
      - Option 2 SSD NAND FTL: 4 GB

Based on the stated assumptions…
What to do with extra DRAM?

- Deploy 1 EB of storage using ~78% less DRAM
- Deploy ~4.5 EB of storage using the same amount of DRAM (~+3.5 EB in same DRAM footprint)
- Deploy 1 EB of storage across ~3,570 servers, each with 64 GB of DRAM.

	Option 1 SSD with Traditional FTL	Option 2 SSD with NAND FTL
Storage System Useable Capacity (TB)	2949.12	2949.12
SSDs in System	24	24
DRAM Required per System (GB)	96 + (24 x 32) = 864	96 + (24 x 4) = 192
Number of systems deployed enabling 1 EB Storage	340	340
DRAM Required for 1 EB Storage (GB)	293,760	65,280

Thank You!

Questions?

